

# Vishal Indivar Kandala

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## RESEARCH PROFILE

Ph.D. Candidate in Mechanical Engineering with a focus on computational science, viewing computation as a third pillar of scientific inquiry alongside theory and experiment. My research centers on developing bespoke, high-fidelity software tools from first principles to investigate complex physical phenomena. My current Ph.D. work involves building **PICurv**, a parallel Eulerian-Lagrangian solver for data-driven turbulence modeling. Previous projects have applied these computational methods to diverse applications, including biomedical fluid-structure interaction in LVADs and physics-informed modeling of cardiac dynamics (EDPVR). This work is driven by a deep appreciation for the predictive power of computational physics, which began with hands-on experience in undergraduate engineering competitions and has grown through tackling progressively complex research challenges.

## EDUCATION

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|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 2021 - present | <b>Ph.D., Mechanical Engineering</b> at Texas A&M University<br>Advisor: Dr. Iman Borazjani<br>Dissertation: Data-Driven Turbulence Closure Models using Filtered Density Function Methods | College Station, TX |
| 2019 - 2021    | <b>M.Eng., Aerospace Engineering</b> at Texas A&M University<br>College Station, TX                                                                                                        | (GPA: 3.8/4.0)      |
| 2014 - 2018    | <b>B.Tech., Mechanical Engineering</b> at VIT University<br>Chennai, India                                                                                                                 | (GPA: 8.0/10.0)     |

## RESEARCH INTERESTS

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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Primary Focus         | Turbulence modeling, Data-driven closure development, Filtered Density Function (FDF) methods, Large Eddy Simulation (LES)         |
| Computational Methods | High-performance computing, Eulerian-Lagrangian methods, Immersed boundary methods, Parallel computing, Numerical methods for PDEs |
| Application Areas     | Turbulent scalar transport, Combustion modeling, Fluid-structure interaction, Biomedical flows, Particle-laden flows               |
| Emerging Interests    | Physics-informed machine learning, Scientific machine learning for turbulence, Hybrid numerical-ML approaches                      |

## RESEARCH EXPERIENCE

**Graduate Research Assistant — Texas A&M University** May 2021 - Present  
*Scientific Computing and Biofluids Lab, Advisor: Dr. Iman Borazjani*

- PICurv: Eulerian-Lagrangian Solver for Turbulent Scalar Transport** Jan 2024 - Present
- Developing a high-fidelity solver for scalar **Filtered Density Function (FDF)** transport within the **CURVIB** immersed boundary framework.
  - Implementing Eulerian-Lagrangian coupling and solving stochastic particle methods (Langevin equation) to model sub-grid scalar fluctuations in turbulent flows.
  - Building a modular, **well-documented**, **open-source tool** with a focus on robust software engineering for the scientific community.

- Optimizing parallel scalability to handle millions of particles across 1,000+ cores using C/C++, MPI, and PETSc.
- This work establishes the foundation for a future [velocity-scalar FDF \(VFDF\)](#) framework, enabling a new class of data-driven turbulence closure models.

## Biomedical Computational Mechanics Projects

May 2021 - Dec 2023

### Left Ventricular Assist Device (LVAD) Flow Analysis:

- Conducted high-resolution, unsteady fluid-structure interaction simulations of pulsatile blood flow within patient-specific LVAD outflow graft geometries.
- Analyzed key hemodynamic metrics including **wall shear stress**, **vorticity**, and regions of flow stasis to quantify the risk of thrombosis (blood clotting) and hemolysis (blood cell damage).
- Utilized the Curvilinear Immersed Boundary (CURVIB) method to accurately capture the interaction between blood flow and the flexible arterial walls.
- Presented a detailed analysis of the outflow graft hemodynamics at the 76th Annual APS Division of Fluid Dynamics meeting.

### Cardiac Pressure-Volume Modeling (EDPVR):

- In a cross-disciplinary collaboration with the [Cardiovascular and Computational Biomechanics Lab](#), developed a novel framework for modeling right ventricular pressure-volume dynamics.
- Designed and implemented a **physics-informed regression model** that extends the established Klotz single-beat method to the anatomically complex right ventricle.
- Generated training data by performing finite element simulations (Abaqus) on a large cohort of biventricular geometries reconstructed from high-resolution MRI scans.
- Demonstrated a **40-60% reduction in prediction error** (nRMSE reduced from 22.7% to 12.3%) by introducing a geometric correction factor, enabling more accurate, non-invasive cardiac assessment. Manuscript in preparation for *Nature Scientific Reports*.

## Undergraduate Research Assistant — VIT University

Jun 2017 - May 2018

Advisor: Dr. [Harish Rajan](#)

### Hypersonic Reentry Vehicle Design Optimization

- Conducted a design optimization study for a modified double-spiked reentry capsule using ANSYS Fluent to mitigate aerothermal loads.
- Performed parameter sweeps varying aerospike frustum geometry, achieving a 25% drag reduction and 16% surface temperature reduction for the optimized configuration.
- Authored a technical report on the findings, demonstrating a systematic approach to design space exploration through computational simulation.

## PUBLICATIONS & PRESENTATIONS

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### Conference Presentations

**Kandala, V.I.** and Borazjani, I.

2025

"A Parallel Particle-In-Cell (PIC) Solver on Curvilinear Grids for Turbulent Flow Simulation," *78th Annual APS Division of Fluid Dynamics*, Houston, TX. (Accepted) [[Schedule](#)]

**Kandala, V.I.** and Borazjani, I.

2023

"High Resolution Numerical Simulations of LVAD Outflow Graft Haemodynamics," *76th Annual APS Division of Fluid Dynamics*, Washington, DC. [[Abstract](#)] [[Presentation](#)]

### Posters

Naeini, V., **Kandala, V.**, et al.

2023

"On the non-invasive estimation of end diastolic pressure volume relationship in the right ventricle," *BMES Annual Meeting*, Seattle, WA. [[Abstract](#)]

**Kandala, V.I.** and Borazjani, I.

2022

"High Performance Computing for Biofluids," *High Performance Research Computing Symposium*, Texas A&M University. [[Poster](#)] [[Event](#)]

### Manuscripts in Preparation

Naeini, V., **Kandala, V.I.**, et al.

"In-silico-generated universal end-diastolic pressure-volume relationship for the right ventricle," *in preparation for Nature Scientific Reports*.

### Undergraduate Publications

**Kandala, V.I.**, Singh, R. and Rajan, H.

2018

"Numerical Study of a Modified Double-Spiked Re-entry Capsule," *IJAE* [[PDF](#)]

## TEACHING EXPERIENCE

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### Teaching Assistant — Texas A&M University

Mar 2020 - May 2025

#### STAT 201: Statistical Methods

Spring 2020 - Spring 2023

- Instructed 200+ students from diverse academic backgrounds in statistical inference
- Developed materials emphasizing practical applications and conceptual understanding

#### MEEN 461: Heat Transfer

Fall 2023

- Taught graduate students and senior undergraduates from ME, AE, and ChemE programs
- Created assignments, projects, and exams; delivered lectures; conducted tutorials using MATLAB and Python

#### MEEN 464: Heat Transfer Laboratory

Fall 2024, Spring 2025

- Supervised small teams (4-5 students) conducting heat transfer experiments
- Developed lab manuals and experimental protocols; managed laboratory equipment

## PROFESSIONAL EXPERIENCE

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### Product Development Intern — EPC Electrical Pvt. Ltd.

Jan 2018 - Aug 2018

- Co-led the end-to-end design, fabrication, and deployment of a novel, ultra-low-cost CNC machine, achieving a time-to-market of under 90 days.
- Reduced customer operating costs by 60% by implementing the GRBL open-source firmware and control platform.

### R&D Engineering Intern — BHEL

Jun 2017 - Aug 2017

- Selected as one of 20 interns from a pool of 300+ applicants to contribute to R&D at India's largest energy equipment producer.
- Developed a MATLAB application to compute heat transfer coefficients in industrial condensers, eliminating commercial software licenses and saving an estimated \$2,500 annually.

## INDEPENDENT PROJECTS

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### Cantera-Radcal: Chemical Kinetics with Radiation

Mar 2020 - Jan 2021

- Extended the open-source [Cantera package](#) to include radiation modeling and higher-order reverse diffusion as part of my master's thesis project.
- Integrated the modified package with a compressible OpenFOAM solver to simulate non-premixed flames, a critical step for high-speed combustion applications.

### Foundational CFD Solver Implementations

Coursework

Self-Study

- Developed a portfolio of solvers from first principles to build a fundamental understanding of numerical methods and solver architecture.
- **Parallel Diffusion Solver:** A 2D Finite Difference solver in C using the PETSc library for parallel execution and scalability analysis. [[GitHub](#)]
- **Artificial Compressibility Solver:** A 2D Navier-Stokes solver for backward-facing step flow using an explicit Runge-Kutta time-marching scheme on a stretched grid. [[GitHub](#)]
- **SIMPLE Algorithm Solver:** A 2D Finite Volume solver in Python implementing the SIMPLE algorithm on a staggered grid for channel and lid-driven cavity flows. [[GitHub](#)]

## OPEN SOURCE SOFTWARE

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### PICurv

[GitHub](#) — [Documentation](#)

High-fidelity Eulerian-Lagrangian solver for particle-laden turbulent flows on curvilinear grids. Features modular design, comprehensive Doxygen documentation, automated test cases, and parallel scalability.

### Cantera-Radcal

[GitHub](#)

Extension of Cantera chemical kinetics package with radiation modeling (RADCAL) and higher-order diffusion. Docker implementation for reproducibility.

### Parallel Diffusion Equation Solver

[GitHub](#)

2D diffusion equation solver using finite difference method with PETSc library's C interface. Features Krylov subspace methods for efficient parallel solution.

### Py-SIMPLE

[GitHub](#)

2D finite volume method solver for incompressible, steady-state flows written in Python with NumPy. Implements SIMPLE algorithm for pressure-velocity coupling.

### Parallel Artificial Compressibility Solver

[GitHub](#)

High-performance CFD solver for 2D steady-state incompressible Navier-Stokes equations on generalized curvilinear grids using artificial Compressibility method.

### 1D Composite Diffusion Solver

[GitHub](#)

Simulates 1D heat conduction in spherical bodies with multiple layers of different material compositions and varying thicknesses in the radial direction.

## TECHNICAL SKILLS

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| Programming & HPC        | C/C++, Python, MATLAB, MPI, OpenMP, PETSc, CUDA (exposure)                                                                                                          |
| CFD & Simulation         | ANSYS Fluent, OpenFOAM, STAR-CCM+ (exposure), Converge (exposure), Cantera                                                                                          |
| Numerical Methods        | Finite Volume Method, Finite Element Method, Immersed Boundary Methods, Spectral Methods, Time Integration (RK, fractional-step), Iterative Solvers (GMRES, Krylov) |
| Meshing & Pre-processing | ICEM CFD, Pointwise, Gmsh, ANSYS Meshing                                                                                                                            |
| Post-processing          | Tecplot, ParaView, EnSight, Matplotlib                                                                                                                              |
| CAD                      | SolidWorks, Creo, SpaceClaim                                                                                                                                        |
| HPC Systems              | National supercomputing resources, cluster management, job scheduling                                                                                               |
| Version Control          | Git, GitHub                                                                                                                                                         |

## AWARDS & HONORS

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|------------|------------------------------------------------------------------------|
| 2022       | <b>Chad Patel Scholarship</b> , American Society of Indian Engineers   |
| 2019, 2020 | <b>Aerospace Academic Excellence Fellowship</b> , Texas A&M University |

## SERVICE & LEADERSHIP

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### Judge — Texas Science Fair

2023

Evaluated student research projects and provided feedback to young researchers

### Department Representative — TAMU Graduate and Professional Student Government

Represented mechanical engineering graduate students in university governance

### Contributing Writer — Ingenium Blog, Texas A&M Engineering

2020 - 2022

Wrote articles about graduate student experiences and research for university engineering blog

### Co-Founder & Head of Aerodynamics — Team Aviators International

2016 - 2017

– Founded first SAE Aero Design team at VIT University; led 5-member aerodynamics sub-team

– Secured \$5,000+ in funding through workshops and sponsorships

### Lead Braking Engineer — Team Saksham (Baja SAE India)

2015 - 2016

– Designed and fabricated braking system; achieved 5th place nationally (100+ teams)

– Contributed to fundraising efforts securing \$3,000+ for components and travel

## RELEVANT COURSEWORK

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### CFD & Turbulence Modeling

Turbulence Modeling (AERO 670), Turbulence Processes (OCEN 640), Computational Fluid Dynamics (MEEN 689)

### Propulsion & High-Speed Flows

High-Speed Combustion & Propulsion (AERO 641), Aerothermochemistry (AERO 676), Advanced Aerodynamics (AERO 601)

### Scientific & Parallel Computing

Scientific Machine Learning (ECEN 689), Parallel Computing (CSCE 689), Advanced Computer-Aided Engineering (MEEN 632)

### Numerical Methods & Analysis

Numerical Heat Transfer & Fluid Flow (MEEN 644), Numerical Methods in Differential Equations (MATH 610), Numerical Analysis (AERO 689)

### Fundamental Mechanics

Theory of Fluid Mechanics (AERO 602), Fluid Mechanics (MEEN 621), Introduction to Random Dynamical Systems (AERO 630)

## PROFESSIONAL MEMBERSHIPS

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American Physical Society (APS) – Division of Fluid Dynamics

## REFERENCES

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### Dr. Iman Borazjani

*Ph.D. Advisor*

Professor, J. Mike Walker '66 Department of Mechanical Engineering

Texas A&M University — Email: [iman@tamu.edu](mailto:iman@tamu.edu)

### Dr. Reza Avazmohammadi

*Research Collaborator*

Associate Professor, Department of Biomedical Engineering

Texas A&M University — Email: [rezaavaz@tamu.edu](mailto:rezaavaz@tamu.edu)

### Dr. Harish Rajan

*Undergraduate Research Advisor*

Associate Professor, School of Mechanical Engineering

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